Samuel W Muwonge

Azure Cloud-DevOps Engineer |

| Cloud (Azure) | AI-Driven Automation | Fintech | Public Sector | Air & Defence | Transportation | Betting | Oil & Gas | Driving Digital Transformation

Enthusiastic DevOps Engineer with expertise in architecting, automating, and optimizing cloud and hybrid infrastructure through Infrastructure as Code, containerization, and CI/CD. Skilled in delivering AI-driven and mission-critical platforms by embedding DevSecOps, observability, and cost optimization into large-scale deployments. Experienced in applying AI/ML Ops practices with tools such as MLflow and LangGraph, and in implementing sovereign cloud AI patterns using Azure Government to meet strict compliance and data residency requirements. Proven track record collaborating across public, aerospace, defence, finance, and manufacturing sectors to enable digital transformation while ensuring scalability, security, compliance, and resilience.

Experience

OCT 2024 - Present
DevOps Engineer | SymphonyAI

An AI SaaS company delivering predictive and generative AI solutions, As a DevOps Engineer at SymphonyAI in the Financial Services vertical, I contribute to building and automating cloud-native platform on Azure. My focus is on enabling secure, scalable, and reliable delivery pipelines and infrastructure to support AI-powered financial analytics, fraud detection, and risk management solutions for enterprise banking clients.

Key Contributions at SymphonyAI (Financial Services vertical):

- On the Fraud Detection Platform, I have completed the design and build of CI/CD pipelines with GitHub Actions and ArgoCD, embedding automated accuracy and compliance checks. This now allows fraud models to be deployed the same day they are approved, giving financial institutions faster protection against emerging threats.
- For the Financial Risk Analytics Engine, I have provisioned hybrid infrastructure with Terraform and Ansible, applying resource tagging and automated scaling policies across Azure and on-prem clusters. This has given finance teams visibility of GPU/CPU usage and prevented unnecessary spend, while still supporting large-scale training workloads.
- On the Regulatory Monitoring Platform, I am finalising the integration of Prometheus, Grafana, and OpenTelemetry to monitor latency, drift, and transaction processing. This is enabling compliance officers to access auditable dashboards and giving data science teams real-time insight into model performance in production.
- On the Client Onboarding Automation Platform, I am in the process of completing automation of environment provisioning with Terraform and Ansible, replacing manual steps with a consistent, compliant approach. Once finalised, this will accelerate onboarding timelines and allow product teams to deliver solutions to new customers more quickly.
- On the AI Model Lifecycle Optimisation Initiative, I am implementing MLflow and LangGraph within CI/CD workflows to track experiments, manage model versions, and automate deployments. This ensures financial clients can reliably deploy and monitor AI models used for fraud detection and risk analysis.
- I am also designing patterns for sovereign AI deployments on Azure Government, ensuring financial services workloads remain compliant with data residency and regulatory obligations while still leveraging scalable cloud infrastructure.

JLY 2023 - OCT 2024
DevOps Engineer | Airbus Defence And Space

At Airbus Defence and Space, a global leader in space and defence innovation, I deliverd secure, automated, and scalable DevOps solutions to support mission-critical defence and space platforms, ensuring reliability, compliance, and faster delivery for government and aerospace stakeholders..

Key Contributions to Airbus Defence and Space:

- Within the Satellite Communications Modernisation Programme, I was tasked with building sovereign cloud landing zones that could meet NATO security requirements. Using Terraform and Bicep, I provisioned Azure Sovereign Landing Zones with AKS clusters and VNet peering to integrate with on-prem Kubernetes. This provided a compliant, scalable environment for deploying classified applications.
- On the Defence Command Platform, security teams were relying on manual checks that slowed delivery. I embedded OPA for policy-as-code and Falco for runtime detection directly into the CI/CD pipeline. This meant only compliant workloads were deployed, streamlining audits and reducing the risk of non-compliant code reaching production.
- For the Mission Control Operations Centre, operations lacked end-to-end observability, which caused delays in diagnosing issues. I implemented Prometheus, Grafana, and ELK stacks with customised dashboards and alerts. This gave 24/7 visibility across systems, enabling proactive incident response and uninterrupted command and control operations.
- Within the Secure Data Exchange Project, sensitive mission data had to be transferred between sovereign cloud and on-prem systems under strict compliance. I designed CI/CD workflows with GitHub Actions to automate encrypted deployments and implemented secrets management with HashiCorp Vault. This ensured secure data transfer pipelines and gave assurance to stakeholders managing classified defence operations.
- Created IaC modules for the AI-Enhanced Mission Analytics Programme using MLflow to manage AI model lifecycles and Azure Government patterns to deploy AI-driven defence applications in line with sovereignty requirements.

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Nationality

BRITISH

Core DevOps Tools

- **Cloud Platforms:**
  Azure
  AWS
- **Containerization & Orchestration:**
  Docker, Kubernetes (AKS), OpenShift, Helm
- **Infrastructure as Code (IaC):**
  Terraform, Azure Resource Manager (ARM) templates
- **CI/CD :**
  Azure DevOps Pipeline, Jenkins, GitHub Actions, GitLab CI/CD, CircleCI, Travis CI, ArgoCD
- **Configuration Management:**
  Ansible, Chef
- **Version Control & Collaboration:**
  Git, GitHub, GitLab, Bitbucket, Azure Repos
- **Monitoring & Logging:**
  Azure Monitor, Log Analytics, Prometheus, Grafana, ELK/EFK Stack, Datadog, Splunk, New Relic

Artifact & Package Management:
Nexus, Artifactory, Harbor

- **Security/DevSecOps:**
  Azure Key Vault, Azure Security Center/ Defender, HashiCorp Vault, Snyk, Aqua Security, SonarQube
- **Scripting | Programming languages:**
  Python, BASH, Groovy, YAML, Powershell, Go
- **Collaboration & Project Management:**
  Azure Boards, Jira, Confluence, Azure DevOps, Slack, Microsoft Teams

Certifications

- Azure DevOps Engineer
- Azure Administrator Associate
- CKA (Certified Kubernete Administrator)
- Docker Certified Associate
- HashiCorp Terraform Associate

Enterprise Trained

- Azure Security Engineer
- Prometheus and Grafana
- Atlassian JIRA Administration
- Certified Scrum Master (CSM)

Agile & Scrum Ways of Working

- **Participating in daily stand-ups** to share progress, highlight blockers, and align with development, QA, and product teams.
- **Collaborating with stakeholders** product owners, business analysts, security, compliance, operations) to refine requirements and ensure technical delivery meets business needs.
- **Contributing to sprint planning** estimating DevOps-related tasks (pipeline improvements, infra changes, automation work) and ensuring they align with development priorities.
- **Showcasing work during sprint reviews/demo days** e.g., demonstrating new CI/CD automation, monitoring dashboards, or infrastructure-as-code deployments to technical and non-technical stakeholders.
- **Retrospectives participation** raising feedback on tooling, processes, or bottlenecks and suggesting improvements to delivery pipelines and team workflows.

APR 2022 - JUL 2023
DevOps Engineer | DWP Digital

DWP Digital is the digital arm of the UK's largest public service department, the Department for Work and Pensions. As a DevOps Engineer at DWP Digital, I contributed to the design, automation, and operation of Azure-based platforms supporting welfare, pensions, and child maintenance services. My work focused on improving delivery speed, system reliability, and compliance, ensuring resilient digital services for millions of UK citizens.

Key Contributions to DWP Digital Services:

- On the Universal Credit platform, manual deployments were causing delays that impacted feature releases for citizens. I implemented CI/CD pipelines with GitHub Actions, embedding automated testing and gated approvals. This reduced deployment effort from hours to minutes and ensured benefits services were continuously available.
- For the Pension Claims Processing System, legacy infrastructure was failing to scale during peak demand periods. I migrated applications into Docker containers and orchestrated them on Kubernetes clusters. This provided seamless scaling during seasonal peaks and improved overall system reliability.
- Within the Citizen Welfare Payments Service, outages were difficult to detect and resolve quickly. I deployed Prometheus, Grafana, and ELK stacks, building dashboards for transaction latency and system health. This gave operations teams the visibility needed to resolve issues faster and ensured uninterrupted access to welfare payments for millions of citizens.
- Integrated MLflow into DevOps pipelines for the Fraud Prevention Pilot in Welfare Claims, enabling model tracking, while also deploying containerised AI services in Azure Government landing zones to meet UK government compliance standards.
- Built and maintained Kafka-based messaging pipelines to support high-volume, event-driven communication between welfare and pension systems, ensuring reliable, real-time processing of citizen data across multiple services.

MAY 2020 - APR 2022
DevOps Engineer | MONZO

The UK's largest digital bank, serving millions of customers through a mobile-first platform. As a DevOps Engineer at Monzo, I focused on building secure, automated, and scalable Azure-based platforms to support core banking, payments, and data services. My work enabled high availability, cost efficiency, and compliance in a regulated environment, ensuring reliable digital banking experiences for millions of customers.

Key Contributions to Monzo Digital Bank:

- Implementation of auto-scaling and resilience patterns on Azure (AKS, Azure SQL Database, Blob Storage, Application Gateway/Load Balancer), improving service availability and reducing customer-impacting incidents, ensuring uninterrupted access to banking services.
- Collaborated with Data Engineering teams to automate data pipeline infrastructure using Terraform, Azure Data Factory, and Event Hubs, cutting provisioning time and enabling analysts and product teams to process terabytes of transaction data daily.
- Supported cloud cost optimisation efforts by building automated Azure Cost Management dashboards and applying right-sizing policies, while improving financial transparency for engineering and finance stakeholders.
- Embeded secrets management and compliance automation with Azure Key Vault, Azure AD, and Vault integrations in delivery pipelines, reducing security risks related to credential exposure and meeting FCA and PSD2 compliance requirements.

SEPT 2018 - MAY 2020
DevOps Engineer | The Game Company

As a DevOps Engineer at The Game Company (TGC), a pioneer in AI-powered cloud gaming and Web3 innovation, I supported the design and automation of low-latency, scalable platforms enabling seamless access to AAA titles worldwide. My contributions focused on optimizing cloud infrastructure, streamlining deployments, and embedding security, ensuring high-performance gaming experiences for players and partners across a global ecosystem.

Key Contributions to The Game Company (TGC):

- Contributed to automating global infrastructure provisioning on Azure and AWS with Terraform and Ansible, cutting environment setup time by 65% and supporting rapid deployment of gaming nodes across Europe, Asia, and North America.
- Helped deploy and optimize containerized game-streaming workloads on Kubernetes (AKS/EKS), reducing latency by 30% and achieving 99.9% uptime, ensuring smooth, lag-free experiences for players streaming AAA titles.
- Supported development teams by building CI/CD pipelines with GitHub Actions and ArgoCD, automating deployments for AI-driven personalization services, reducing release errors by 35% and speeding up feature delivery.
- Worked with blockchain engineers to integrate Web3 services (wallets, NFT marketplaces, multi-chain APIs) into pipelines, improving release frequency by 40% and enhancing digital ownership features for players and partners.
- Assisted in introducing cloud cost monitoring and optimisation practices with Azure Cost Management and FinOps dashboards, lowering infrastructure spend by 18% while maintaining performance SLAs and providing cost transparency to stakeholders.
- Built and maintained Kafka-based event streaming pipelines to handle real-time player interactions, game telemetry, and matchmaking data across distributed cloud environments, ensuring low-latency performance and reliability at global scale.

Career Journey into DevOps

I began my career with a strong interest in how technology could transform the way organisations deliver services. Early on, I worked in roles that exposed me to infrastructure, cloud, and automation, where I quickly saw the bottlenecks teams faced with manual deployments and slow delivery cycles.

Curiosity led me deeper into scripting, cloud platforms, and infrastructure-as-code skills that naturally aligned with DevOps practices. I started contributing to CI/CD pipelines, automating provisioning, and supporting cloud migrations, and I found real satisfaction in seeing how automation reduced friction for developers and improved outcomes for end users.

Over time, I gained hands-on experience across industries from banking and AI SaaS to defence, gaming, and public services. Each role gave me a broader perspective: how regulated industries value security and compliance, how high-traffic platforms need reliability and scalability, and how data-driven organisations rely on observability and automation to innovate.

In more recent roles, I have extended this journey into AI/ML Ops, applying tools such as MLflow and LangGraph to manage the lifecycle of AI models, integrate them into CI/CD pipelines, and ensure reliability in production. I have also contributed to projects requiring sovereign cloud AI patterns using Azure Government, where compliance and data residency are paramount for regulated industries such as defence and financial services.

Today, as a DevOps Engineer, I not only rely on core technologies like Azure, AWS, Kubernetes, Terraform, and CI/CD pipelines, but I also embrace AI-driven tools and workflows that significantly boost efficiency. From accelerating troubleshooting with AI copilots to automating documentation, testing, and monitoring, AI has become an integral part of how I deliver value faster and smarter.

For me, DevOps is not just a role but a mindset continuously learning, improving, and evolving with AI to build platforms that empower teams to innovate and deliver at scale.

May 2017 - Sept 2018

Cloud Engineer | ABB

ABB is A global leader in electrification, robotics, automation, and digital industries, delivering sustainable technology solutions across more than 100 countries. at ABB UK, I supported the delivery of secure and scalable cloud solutions across Azure. My role focused on automating infrastructure, improving reliability, and enabling digital platforms that powered ABB's electrification, robotics, and industrial automation services.

Key Contributions to ABB Digital Sevices:

- Migration of legacy industrial applications to Azure, reducing infrastructure costs by 20% and improving scalability for ABB's electrification and robotics business units.
- Automated infrastructure provisioning and configuration with Terraform, Ansible, and Azure Resource Manager (ARM/Bicep templates), cutting environment setup time by 60% and enabling faster delivery of digital platforms to engineering and R&D teams.
- Improved cloud performance and reliability by optimising VNet design, load balancing (Azure Load Balancer/Application Gateway), and autoscaling policies, boosting application response times by 35% and enhancing customer experience across ABB's connected solutions.
- Supported monitoring and operational visibility by implementing Azure Monitor and Log Analytics, reducing incident detection and resolution times by 30% and providing service owners with real-time performance insights.
- Enhanced cloud security and compliance posture by deploying Azure AD identity controls, Key Vault, automated policy checks (Azure Policy), and encrypted storage, lowering audit findings by 25% and ensuring alignment with ABB's global security standards.

JLY 2014 - APR 2017

Cloud Engineer | Cargill

One of the world's largest food and agriculture companies, operating across 70+ countries. As a Cloud Engineer at Cargill, I supported the design and automation of AWS based platforms powering food supply, processing, and sustainability initiatives. Focussing on cloud migration, performance optimisation, and security, enabling reliable digital solutions across Cargill's global operations.

Key Contributions to Cargill Supply Chain:

- Supported the migration of supply chain applications to AWS, reducing infrastructure costs by 22% and improving scalability for global logistics and food processing systems.
- Automated infrastructure builds with Terraform cutting provisioning time by 55% and enabling faster delivery of digital solutions to agriculture and sustainability teams.
- Improved system performance and reliability by optimising autoscaling groups, storage tiers, and load balancers, reducing application downtime by 35% and enhancing efficiency in food production and distribution workflows.
- Implemented cloud monitoring and observability using AWS CloudWatch, Azure Monitor, and custom dashboards, reducing mean time to detection (MTTD) and recovery (MTTR) by 30%, giving global operations teams real-time visibility.
- Strengthened cloud security compliance by deploying IAM controls, encryption policies, and automated vulnerability scans, reducing critical security findings by 28% and ensuring alignment with regulatory and corporate standards.

" DevOps Engineer experienced in Cloud, automation, and Agile delivery, with a track record of improving reliability, security, and speed of digital platforms. Open to new opportunities where I can add value, grow my expertise, and help teams deliver at scale "

Samuel Wasswa Muwonge
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